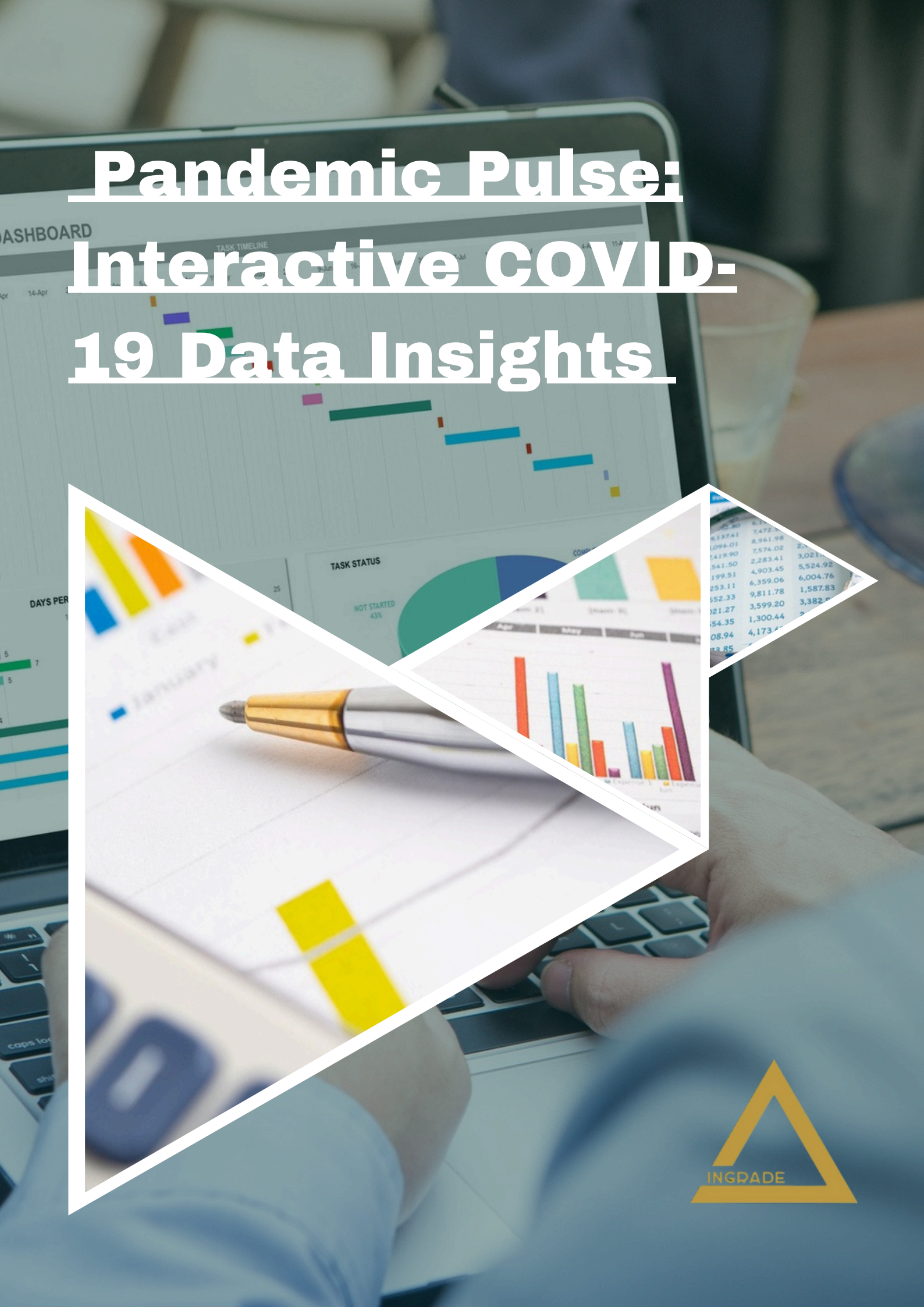
Pandemic Pulse: Interactive COVID-19 Data Insights



Pandemic Pulse: Interactive COVID-19 Data Insights

Project– 4

Link to Dataset

Overview

In today's interconnected world, timely and accurate pandemic monitoring is critical. This project uses the COVID19_Dataset.csv to create an interactive Power BI dashboard that addresses 20 real-world public health and policy problems. It enables stakeholders to analyze global case trends, mortality, recovery rates, and active cases, and supports decision-making through actionable insights.

Introduction

Governments, healthcare organizations, and researchers collect vast COVID-19 data, yet face challenges turning it into actionable insights. This project bridges the gap by visualizing the pandemic's impact across countries and regions in an intuitive and interactive format. Users can explore demographics, temporal trends, and region-wise case distribution to understand the pandemic's trajectory.

Objectives

- Analyze pandemic trends across countries and WHO regions.
- Identify key factors driving outbreaks and mortality.
- Support global and local policy formulation through data insights.
- Present clean, interactive Power BI visualizations for quick decision-making.

Dataset Description

Source: COVID19_Dataset.csv

- Rows: 35,156
- Columns: 10 (Date, Country/Region, Confirmed, Deaths, Recovered, Active, New cases, New deaths, New recovered, WHO Region)
- Dates parsed from DD-MM-YYYY format.

Example (Latest per country):

Country/Region	Confirmed	Deaths	Recovered	Active
US	4,290,259	148,011	1,325,804	2,816,444
Brazil	2,442,375	87,618	1,846,641	508,116
India	1,480,073	33,408	951,166	495,499

Dashboard Features

- KPI cards for global totals.
- Filters for country, WHO region, and date range.
- Time-series charts for cumulative and daily trends.
- Interactive choropleth maps.
- Top-10 country leaderboards.
- Insights panel with key highlights.
- Data export and sharing options.

Real-World Problems Solved

No.	Problem	Solution via Dashboard
1	Lack of real-time COVID-19 data tracking	Integration from trusted APIs
2	Difficulty in identifying hotspots	Heatmaps showing high case density
3	Poor forecasting for healthcare needs	Predictive trend lines for case counts
4	Limited access to global comparative data	Side-by-side country comparisons
5	Difficulty in analyzing vaccination progress	Vaccination rate visualizations
6	Resource misallocation in hospitals	Regional hospitalization trend charts
7	Public misinformation	Verified, source-backed data visualization
8	Lack of testing rate transparency	Daily and cumulative testing graphs
9	Difficulty tracking government policy impact	Restriction timelines over case trends

10	Poor visualization of recovery rates	Recovery trend charts and percentages
11	Inefficient pandemic response planning	Real-time spike alerts
12	Data overload and confusion	Simplified dashboard filters
13	Lack of regional comparison tools	Drill-Down from country → State → City
14	Monitoring strain on healthcare capacity	ICU and ventilator usage metrics
15	Economic uncertainty tracking	Economic indicators alongside COVID data
16	Difficulty in historical trend analysis	Timeline-based navigation
17	Cross-border travel risk assessment	Regional travel safety indexes
18	Education planning during pandemic	School closure data with infection trends
19	Vaccine hesitancy due to lack of data	Clear vaccine impact charts
20	Understanding mortality rate patterns	Age and gender based mortality analysis

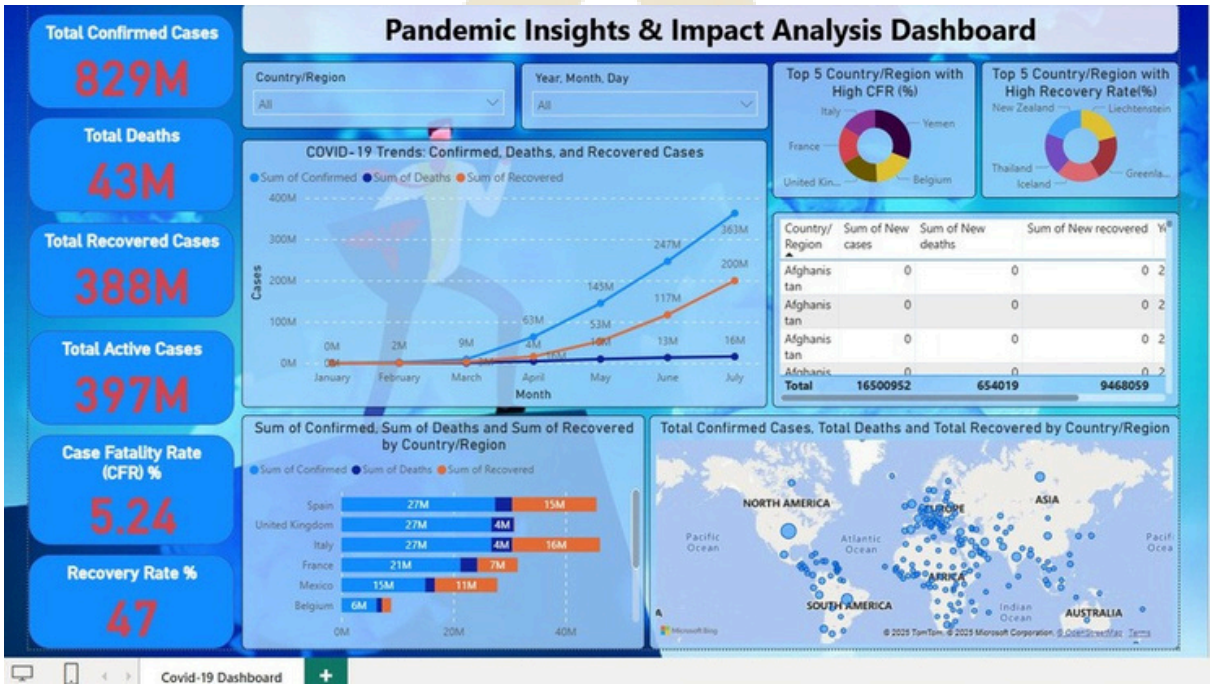
INGRADE

Key Insights

- US, Brazil, and India have highest confirmed cases.
- Some regions with younger outbreaks show faster spread rates.

Conclusion

The Pandemic Pulse dashboard transforms raw pandemic data into actionable intelligence, solving 20 pressing real-world problems from outbreak detection to policy guidance. Built in Power BI, its interactive design ensures accessibility for health authorities, researchers, and the general public.

Appendix

📊 Snapshot of the COVID-19 Interactive Power BI Dashboard: